different activation thresholds and Claude-generated descriptions). However, one feature (6262) has minimal activation on the other two maximally activated examples, demonstrating variations in the *type* of glycosylrtansferase identified and also an asymmetry between 6262 appearing more selective in which glycosyltransferases activate it. While the other two features activate on all examples, but particularly on both respective maximally activating examples, when both compared on a separate protein, we see the specific locations within this protein activated are nearby but focused on different regions.

F Additional steering experiments

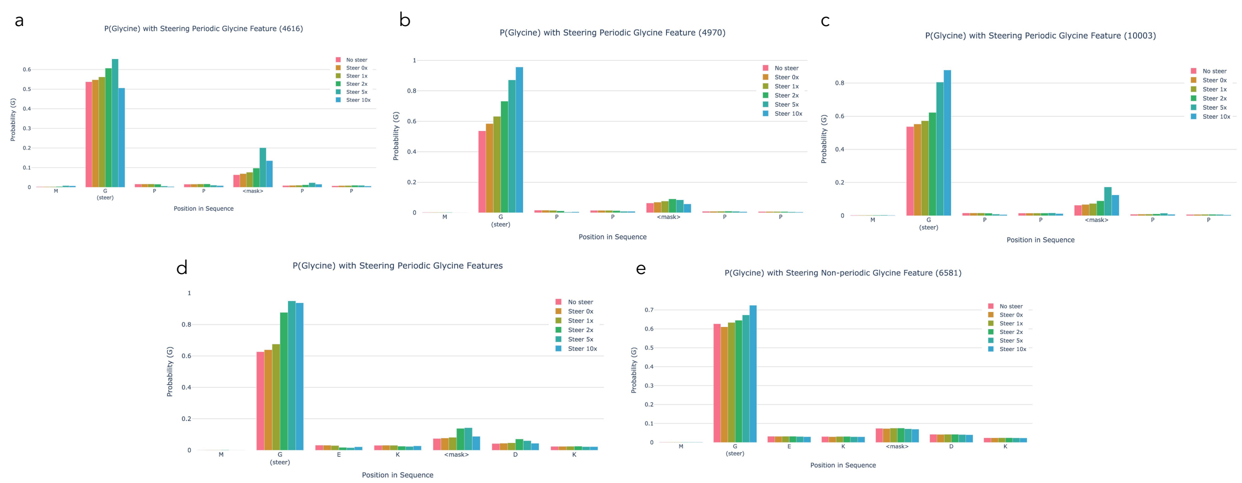

Figure 12: Top row (a-c): Steering MGPP<mask>PP on each of the individual Periodic Glycine Features. Bottom row: Steering an alternate sequence, MGEK<mask>DK. (d) Steering all periodic glycine features (e) Steering non-periodic Glycine feature

In Figure 12, we see that each periodic glycine feature can be used to steer the model to predict a periodic glycine pattern. We also evaluate a variant of the original sequence on the combination of periodic Glycine features and the non-periodic Glycine feature, observing that both increase the probability of Glycine on the steered position. As with the original sequence, we again see that steering has no positive impact on the effect of the Non-periodic Glycine (rather it has a small, negative impact).

Again, in Figure 13, we now test the same 3 periodic Glycine features (4616,4970, 10003), along with the 3 features with highest F1 to Glycine (6581, 781, 5489), and 3 randomly selected features (0, 1,000, 10,000). Here we calculate the slope of p(Glycine) across steering increasing [0, 0.5,0.75, 1, 1.5, 1.75, 2]. Alternate variants of the originally steered sequence maintain similar results and again, we see that even when applying 3 features at a time, the periodic Glycine features have a stronger effect on the masked position's Glycine probability than random or Glycine-specific features. This is also maintained in the longer sequence with multiple the masked periodic patterns being steered until this effect diminishes around the 5th masked position.

It should be noted with all of these experiments that we have only tested a few, somewhat constrained sequences so far, and more work needs to be done to evaluate the contexts in which steering this feature can or cannot work.

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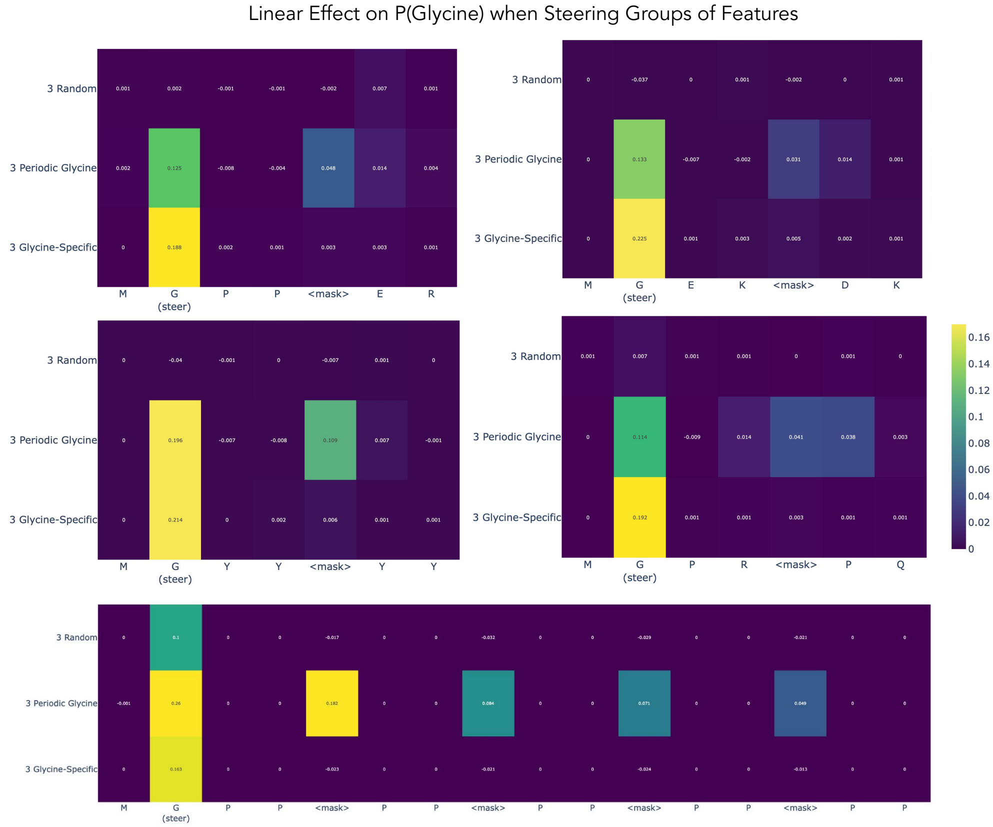

Figure 13: Measuring the linear effect on the predicted probability of Glycine. Evaluated p(Glycine) at each position in a sequence as groups of features were steered on first Glycine position at varying levels (0, 0.5,0.75, 1, 1.5, 1.75, 2). Calculated slope of p(Glycine) with respect to steering amount and visualize this for four sequences across each group of features. 3 Random features (f/0, f/1000, f/10000) were selected based on no activation-based information, 3 Periodic Glycine features (f/4616, f/4970, f/10003) selected based on maximum activation on periodic glycine patterns in collagen, and 3 Glycine-Specific features (f/6581, f/781, f/5389) selected to have the highest F1 scores to Glycine.

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